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Computer Networking

Lab 01: Wireshark Getting Started

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A. Overview

1.1 Goals

- Familiarize yourself with **Wireshark** software - a popular packet capture tool, for research on the operation of network layers.

1.2 Knowledge base

- Basic knowledge of computer networks, operating principles of computer networks.

1.3 Experimental environment

- Students need to prepare in advance a computer/laptop running Window/Linux/MacOS with Internet connection and install the following software:

Wireshark software

This is a packet capture software in the network (sniffing) to help monitor the process of data exchange in the network. For this course, Wireshark helps to analyze and better understand the operation of the layers of the network.

You can download it here: https://www.wireshark.org/download.html

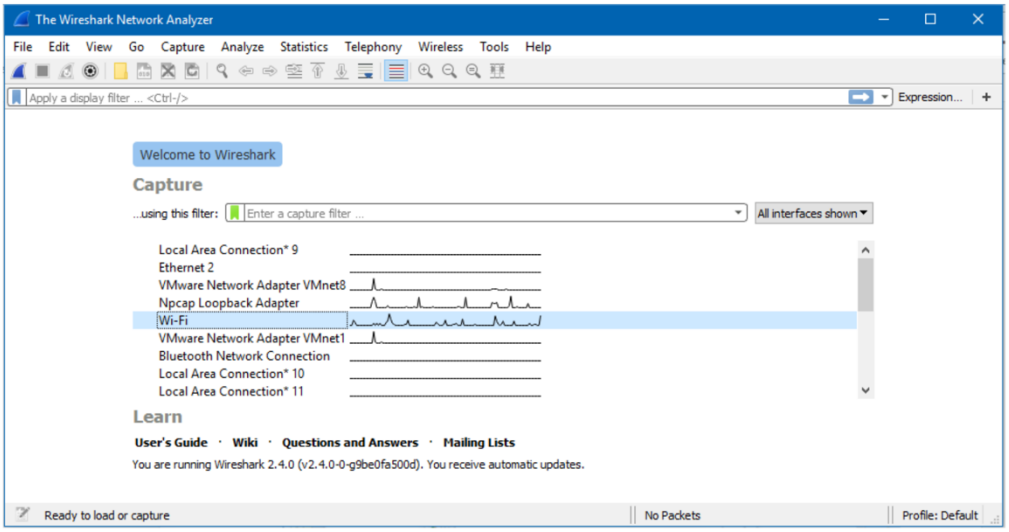


Figure 1. Main UI of Wireshark

B. Practice

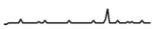
1. Task 1: Getting started with Computer network

Before starting the practice, students should answer the following questions:

- *Name the types of Network-related devices that you know or are using (with illustrations).*
- *What problems can happen if there is no Internet connection for 5 minutes?*
- *Your knowledge goal after completing the Computer Networking course is what?*

2. Task 2: Getting familiar with Wireshark and practice capturing network packets

2.1 Wireshark

- **Wireshark** is a very popular and free packet sniffer that runs on Windows, Linux, MacOS, supports packet capture and observes the content of messages exchanged by protocols at network layers. different.
- In addition, Wireshark also serves to investigate digital evidence (forensic) related to computer network cases.
- The main interface when opening Wireshark will look like *Figure 1*.
- In the first window, you will see a list of network cards (or *network interfaces*) in the **Capture** section. Observe to the right of the interface name will have an illustration showing data exchange activities in the network, when there are signs like  , it can be determined that there is data being exchanged through that interface.
- Depending on the connection type and operating system in use, the names of the interfaces will be different. For example, the names of common interfaces on Windows are as follows:

- Wired connection: Ethernet, Local Area Connection (LAN).
- Wireless connection: WiFi.

- Then, the packet capture interface will appear as follows.

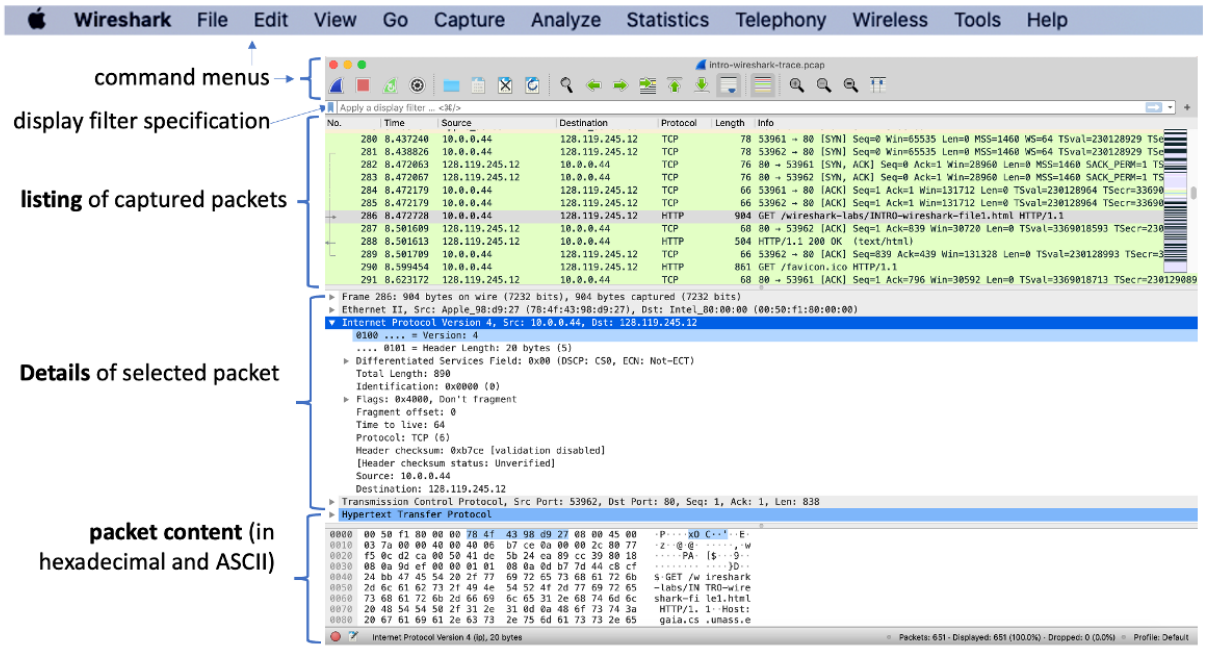


Figure 2. Packet capturing view of wireshark

Note: To be able to capture packets going through it, the network card needs to be enabled in **Promiscuous** mode *(by default it will be enabled in Wireshark).*

The Wireshark interface has five major components:

- The **command menus** are standard pulldown menus located at the top of the Wireshark window (and on a Mac at the top of the screen as well; the screenshot in *Figure 3* is from a Mac). Of interest to us now are the File and Capture menus. The File menu allows you to save captured packet data or open a file containing previously captured packet data and exit the Wireshark application. The Capture menu allows you to begin packet capture.

- The **packet-listing window** displays a one-line summary for each packet captured, including the packet number (assigned by Wireshark; note that this is not a packet number contained in any protocol's header), the time at which the packet was captured, the packet's source and destination addresses, the protocol type, and protocol-specific information contained in the packet. The packet listing can be sorted according to any of these categories by clicking on a column name. The protocol type field lists the highest-level protocol that sent or received this packet, i.e., the protocol that is the source or ultimate sink for this packet.
- The **packet-header details window** provides details about the packet selected (highlighted) in the packet-listing window. (To select a packet in the packet-listing window, place the cursor over the packet's one-line summary in the packet-listing window and click with the left mouse button.). These details include information about the Ethernet frame (assuming the packet was sent/received over an Ethernet interface) and IP datagram that contains this packet. The amount of Ethernet and IP-layer detail displayed can be expanded or minimized by clicking on the plus/minus boxes or right/downward-pointing triangles to the left of the Ethernet frame or IP datagram line in the packet details window. If the packet has been carried over TCP or UDP, TCP or UDP details will also be displayed, which can similarly be expanded or minimized. Finally, details about the highest-level protocol that sent or received this packet are also provided.

- The **packet-contents window** displays the entire contents of the captured frame, in both ASCII and hexadecimal format.
- Towards the top of the Wireshark graphical user interface, is the **packet display filter field,** into which a protocol name or other information can be entered in order to filter the information displayed in the packet-listing window (and hence the packet-header and packet-contents windows). In the example below, we'll use the packet-display filter field to have Wireshark hide (not display) packets except those that correspond to HTTP messages.

2.2 Taking Wireshark for a Test Run

The best way to learn about any new piece of software is to try it out! We'll assume that your computer is connected to the Internet via a wired Ethernet interface or a wireless 802.11 WiFi interface. Do the following:

1. Start up your favorite web browser, which will display your selected homepage.
2. Start up the Wireshark software. You will initially see a window similar to that shown in *Figure 2.* Wireshark has not yet begun capturing packets.
3. To begin packet capture, select the Capture pull down menu and select *Interfaces.* This will cause the "Wireshark: Capture Interfaces" window to be displayed (on a PC) or you can choose Options on a Mac. You should see a list of interfaces, as shown in *Figure 3* (Windows) and *Figure 4* (Mac).

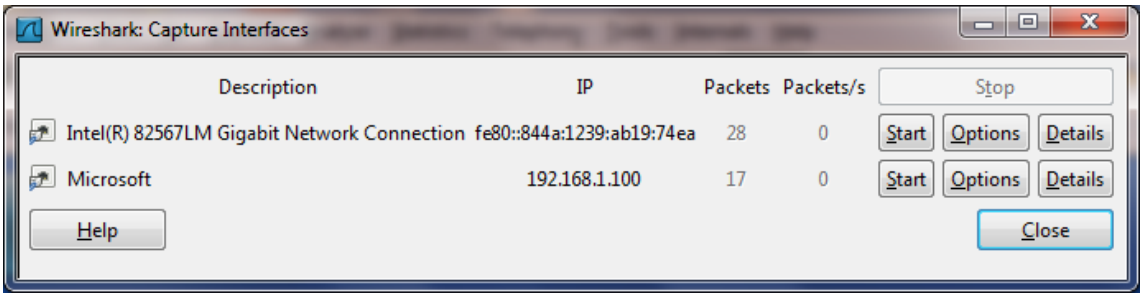


Figure 3. Wireshark Capture interface window, on a Windows computer

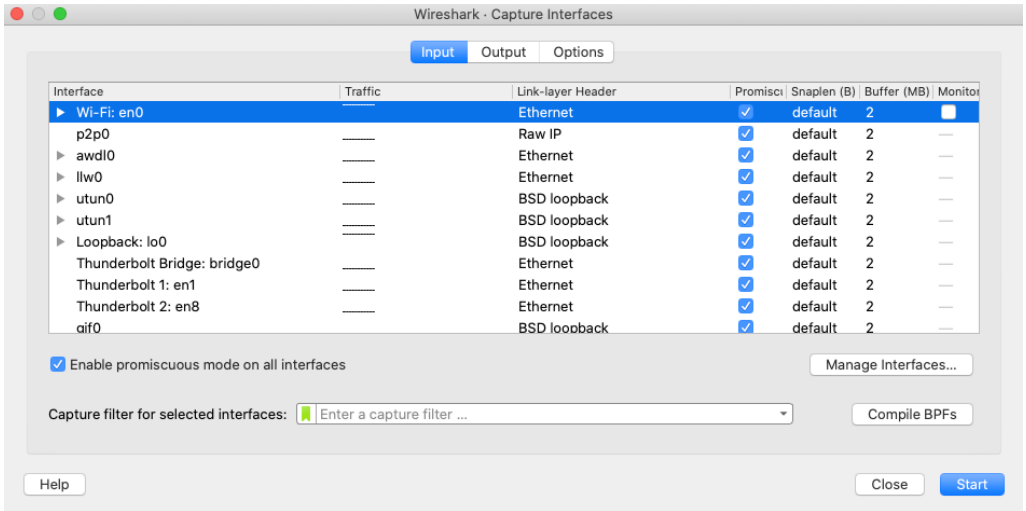


Figure 4. Wireshark Capture interface window, on a Mac computer

4. You'll see a list of the interfaces on your computer as well as a count of the packets that have been observed on that interface so far. On a Windows machine, click on *Start* for the interface on which you want to begin packet capture (in the case in *Figure 3*, the Gigabit network Connection). On a Windows machine, select the interface and click Start on the bottom of the window). Packet capture will now begin - Wireshark is now capturing all packets being sent/received from/by your computer!
5. Once you begin packet capture, a window similar to that shown in *Figure 2* will appear. This window shows the packets being captured. By selecting

Capture pulldown menu and selecting *Stop*, or by click on the red Stop square, you can stop packet capture. But don't stop packet capture yet. Let's capture some interesting packets first. To do so, we'll need to generate some network traffic. Let's do so using a web browser, which will use the HTTP protocol that we will study in detail in class to download content from a website.

6. While Wireshark is running, enter the URL: http://gaia.cs.umass.edu/wireshark-labs/INTRO-wireshark-file1.html and have that page displayed in your browser. In order to display this page, your browser will contact the HTTP server at gaia.cs.umass.edu and exchange HTTP messages with the server in order to download this page, as discussed in section 2.2 of the text. The Ethernet or WiFi frames containing these HTTP messages (as well as all other frames passing through your Ethernet or WiFi adapter) will be captured by Wireshark.
7. After your browser has displayed the INTRO-wireshark-file1.html page (it is a simple one line of congratulations), stop Wireshark packet capture by selecting stop in the Wireshark capture window. The main Wireshark window should now look similar to *Figure 2*. You now have live packet data that contains all protocol messages exchanged between your computer and other network entities! The HTTP message exchanges with the gaia.cs.umass.edu web server should appear somewhere in the listing of packets captured. But there will be many other types of packets displayed as well (see, e.g., the many different protocol types shown in the *Protocol* column in *Figure 2*). Even though the only action you took was to download

a web page, there were evidently many other protocols running on your computer that are unseen by the user. We'll learn much more about these protocols as we progress through the text! For now, you should just be aware that there is often much more going on than "meet's the eye"!

8. Type in "http" (without the quotes, and *in lower case* – all protocol names are in lower case in Wireshark, and make sure to press your enter/return key) into the display filter specification window at the top of the main Wireshark window. Then select *Apply* (to the right of where you entered "http") or just hit return. This will cause only HTTP message to be displayed in the packet-listing window. Figure 5 below shows a screenshot after the http filter has been applied to the packet capture window shown earlier in *Figure 3.* Note also that in the Selected packet details window, we've chosen to show detailed content for the Hypertext Transfer Protocol application message that was found within the TCP segment, that was inside the IPv4 datagram that was inside the Ethernet II (WiFi) frame. Focusing on content at a specific message, segment, datagram and frame level lets us focus on just what we want to look at (in this case HTTP messages).

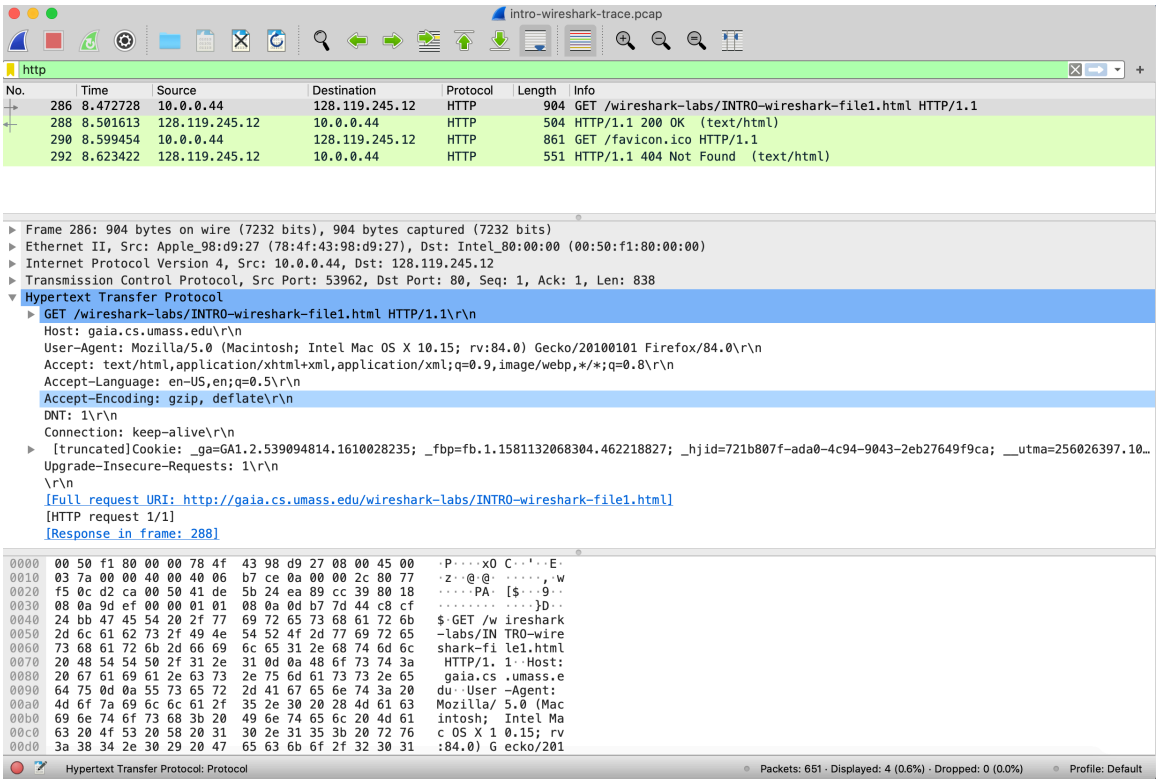


Figure 5. Looking at the details of the HTTP message that contained a GET of http://gaia.cs.umass.edu/wireshark-labs/INTRO-wireshark-file1.html

9. Save Wireshark file captured to .pcapng file with format *ID-Excercise1.pcapng*. Example: *20520001-Excercise1.pcapng*.
10. Find the HTTP GET message that was sent from your computer to the gaia.cs.umass.edu HTTP server. (Look for an HTTP GET message in the "listing of captured packets" portion of the Wireshark window (see Figures 2 and 5) that shows "GET" followed by the gaia.cs.umass.edu URL that you entered. When you select the HTTP GET message, the Ethernet frame, IP datagram, TCP segment, and HTTP message header information will be

displayed in the packet-header window[1]. By clicking on '+' and '-' and right-pointing and down-pointing arrowheads to the left side of the packet details window, *minimize* the amount of Frame, Ethernet, Internet Protocol, and Transmission Control Protocol information displayed. *Maximize* the amount information displayed about the HTTP protocol. Your Wireshark display should now look roughly as shown in *Figure 5*. (Note, in particular, the minimized amount of protocol information for all protocols except HTTP, and the maximized amount of protocol information for HTTP in the packet-header window).

11. Choose 1 website, which you usually access, for example *uit.edu.vn*, *tinhte.vn*,... and capture on it.
12. Save Wireshark file captured to .pcapng file with format *ID-Excercise2.pcapng*.
13. Exit Wireshark.

2.3 Analyze packet capture results from Wireshark

Students perform the practical steps as instructed in section 2.2 to get 2 pcapng output files from Wireshark. Open each file corresponding to the two websites in turn and answer the following questions:

Note: *Present the answer with a screenshot corresponding to the location of the observed information on Wireshark.*

[1] Recall that the HTTP GET message that is sent to the gaia.cs.umass.edu web server is contained within a TCP segment, which is contained (encapsulated) in an IP datagram, which is encapsulated in an Ethernet frame. If this process of encapsulation isn't quite clear yet, review section 1.5 in the text

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1. What is the total packet capture time in each tested website and the total number of packets captured?

2. Among the captured packets, how many HTTP packets are there in total?

3. List at least 5 different protocols that appear in the protocol column (Protocol) when the "http" filter is not applied when accessing 2 websites. Search the Internet and briefly describe the main function of those protocols.

Note: The HTTP - 200 OK return packet only appears when catching the packet on the first visit on the browser.

4. Identify the first HTTP GET packet sent to the tested website. What is the basic purpose of this packet?

5. Identify the response packet corresponding to the HTTP GET packet in Question 4. What information indicates this?

6. Calculate the time from when the first HTTP GET packet was sent until the HTTP 200 OK response packet was received for the tested website. (By default, the value of the time column (Time) in the packet-listing window is the amount of time in seconds since the Wireshark program started capturing the packet).

7. Content displayed on the website gaia.cs.umass.edu

"Congratulations! You've downloaded the first Wireshark lab file!"

Is it included in the captured HTTP packets? If yes, find and confirm

locate this content in the captured packets.

8. What is the IP address of gaia.cs.umass.edu and the website selected in step 11? What is the computer's IP address in use?

9. Through the above packet capture experiment and the packet capture results from Wireshark, let's briefly describe what happens when you start accessing a link to a website until you can view the content on that website.

Expand: What do you think the IP address is for and is there any other way to see the IP address of a computer and another website? Let's make an illustrative example.